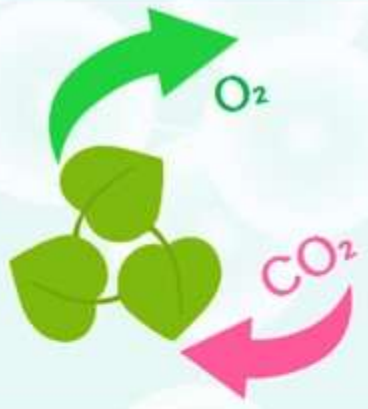


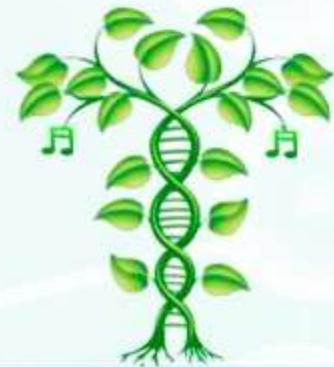


BIOLOGY



LIFE PROCESSES

MODULE - 9



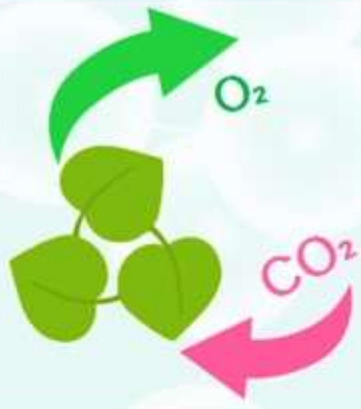
WHAT ARE LIFE PROCESSES?



Life processes are the series of actions that are essential to determine if an animal is alive. Living things have seven essential processes in common:

- Movement,
- Sensitivity,
- Growth and nutrition
- Respiration
- Reproduction and
- Excretion



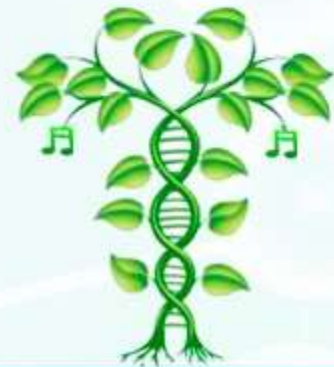


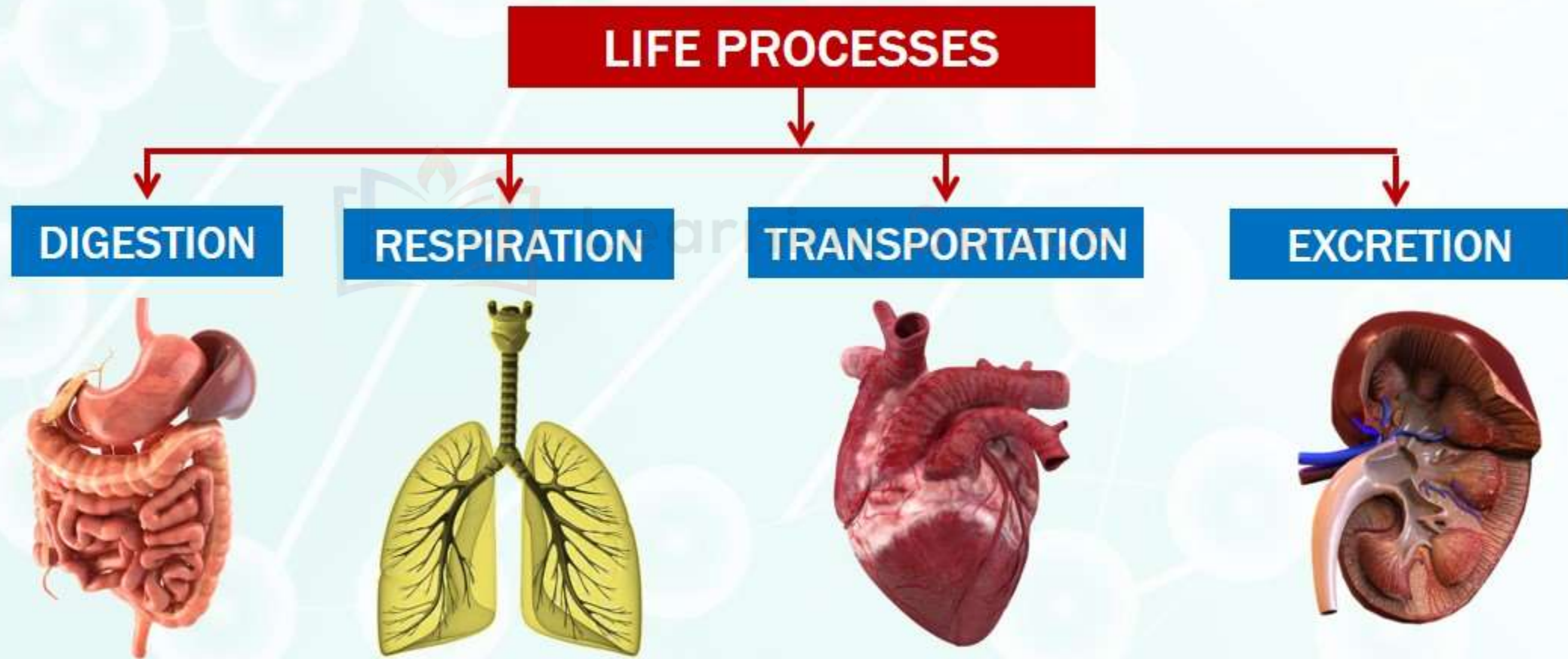
BIOLOGY



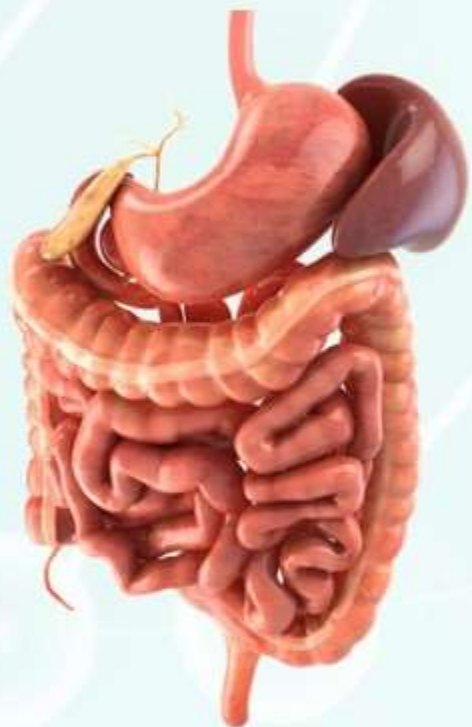
FOOD & NUTRITION

MODULE - 1





DIGESTIVE ORGAN SYSTEM IN HUMANS

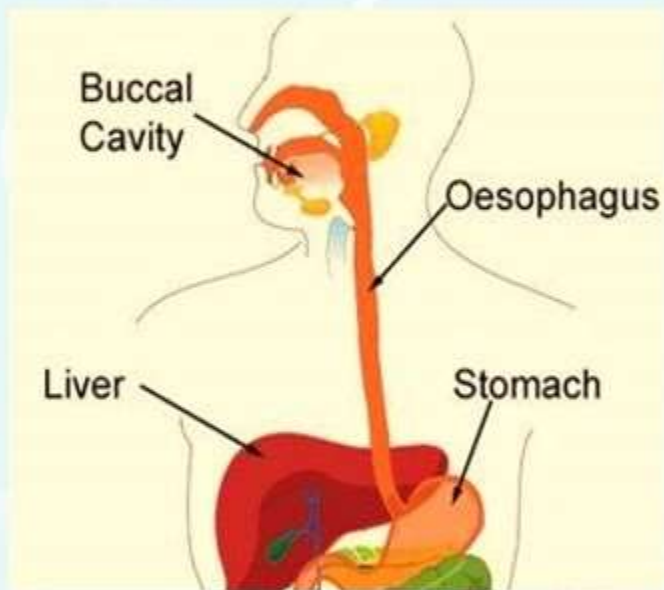


- Human digestion occurs in the organ system of the gastro-intestinal tract /GI Tract / alimentary canal.
- The whole human GI tract is about nine metres (30 feet) long .
- The GI tract releases hormones from enzymes to help regulate the digestive process.
- These hormones are mediated through endocrine or autocrine glands.

Endocrine glands – secretes hormones to target organs
Autocrine glands – secretes hormones for the organ itself



THE ALIMENTARY CANAL

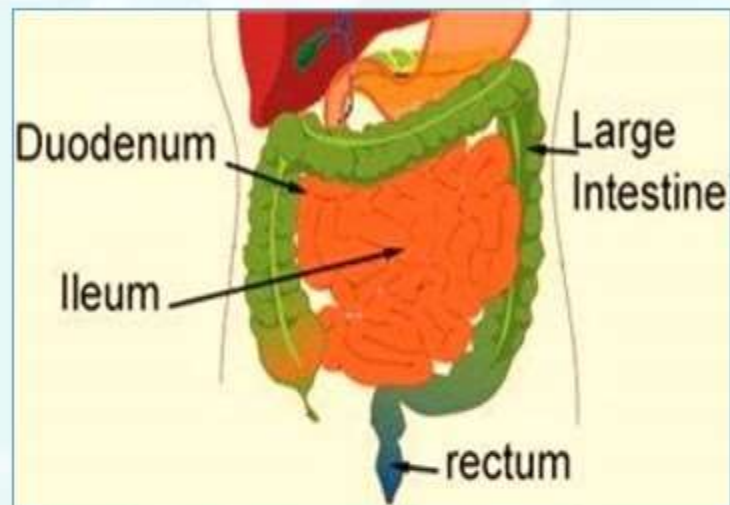


ORGAN	FUNCTION
BUCCAL CAVITY	Food turns into a bolus, Salivary amylase acts on food, chewed food pushed towards pharynx
OESOPHAGUS	facilitates swallowing by mucus secretion from glands, muscular walls promote movement of food, peristaltic movement
LIVER	Produces bile, wide range of other biochemical reactions.
GALL BLADDER	Stores bile
STOMACH	Secretes HCl, kills all harmful micro-organisms, releases enzyme called as pepsin to break down protein, churns food into a chyme

Stomach secrete pepsinogen, an inactive precursor of the enzyme which is activated to pepsin in acidic conditions



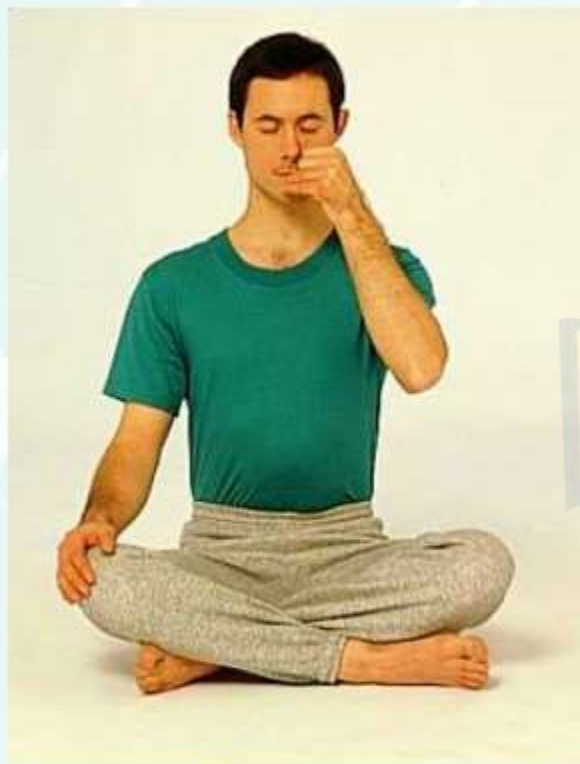
THE ALIMENTARY CANAL



ORGAN	FUNCTION
PANCREAS	Secretes Pancreatic lipase, amylase etc. into the small intestine
DUODENUM	Brunner's glands in it produce mucus, neutralize the acidic chyme, digestion of carbohydrate, proteins and fats.
JEJUNUM	Final conversion of glucose, amino acids and fatty acids by intestinal enzymes, absorption of digested food by enterocytes, involved in magnesium absorption
ILLIUM	Mainly absorption of the digested matter, villi present in the folds for increased surface.
LARGE INTESTINE	removes water from the waste, transports undigested matter from small intestine to rectum
RECTUM	Temporary storage site for fecal matter, ejected through anal canal.



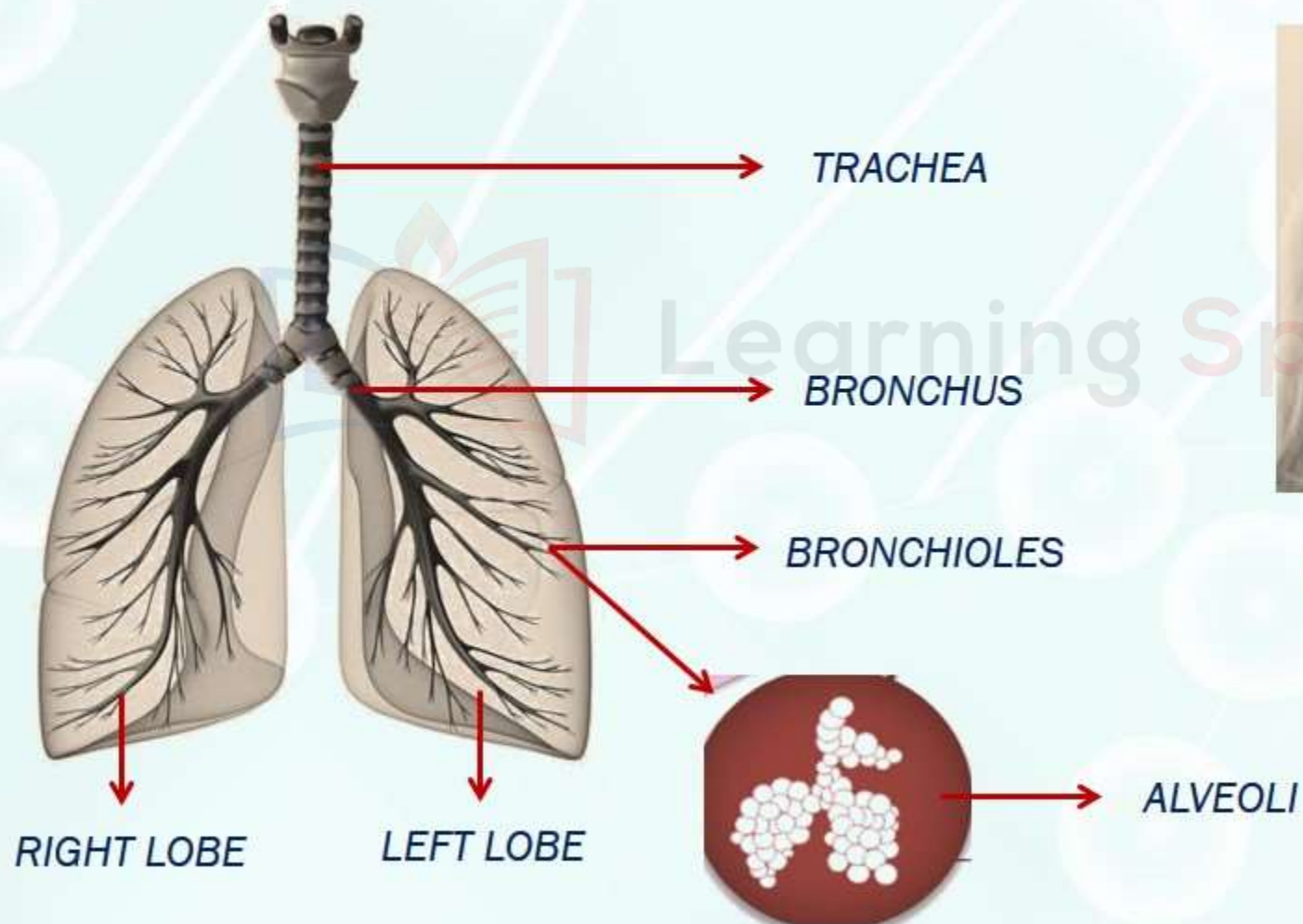
RESPIRATION



- The exchange of oxygen and carbon-dioxide from the atmosphere to the body cells via the process of inhalation and exhalation is called respiration.
- Cellular respiration is the set of metabolic reactions and processes that take place in the cells of an organism to oxidise glucose into ATP; energy for the cell.



RESPIRATORY ORGAN SYSTEM IN HUMANS



THORACIC CAVITY



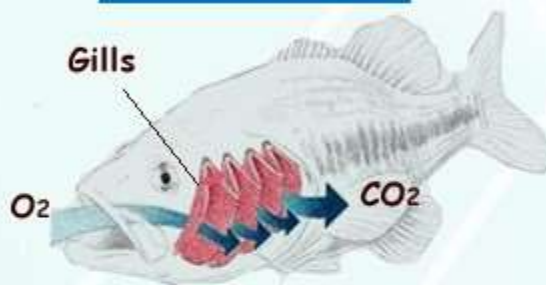
RESPIRATION IN OTHER LIVING BEINGS

SINGLE CELLED ORGANISM



Amoeba respire through surface diffusion of gases

FISH



Gills help the fish to take oxygen from water

INSECTS



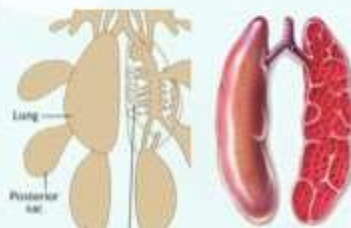
Insects breathe through spiracles which open into tracheae

AMPHIBIANS



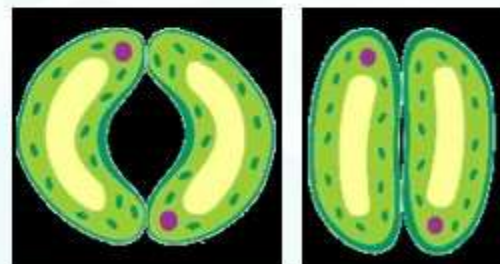
Breathe through lungs in land and through moist skin in water (cutaneous respiration)

BIRDS AND REPTILES



Birds and Reptiles breathe through lungs

PLANTS



Plants exchange their gases during the day when the stomata is open



RESPIRATION

AEROBIC RESPIRATION

- Takes place in presence of Oxygen
- End products – CO_2 and water
- More energy is released.
- Takes place in mitochondria
- Complete oxidation of glucose takes place
- It takes place in
 - most aerobic organisms including humans
 - Aerobic Bacteria

ANAEROBIC RESPIRATION

- Takes place in absence of Oxygen
- End products – Ethanol and CO_2
- Less energy is released
- Takes place only in cytoplasm
- Incomplete oxidation of oxidation takes place.
- It occurs in
 - Anaerobic Bacteria
 - Yeast
 - Human voluntary muscle cells

In yeast, the end product is ethanol

In muscle cells, the end product is lactic acid



RESPIRATION

AEROBIC RESPIRATION

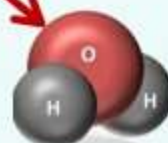
GLUCOSE

PYRUVATE

ATP

ATP

ATP



ANAEROBIC RESPIRATION

GLUCOSE

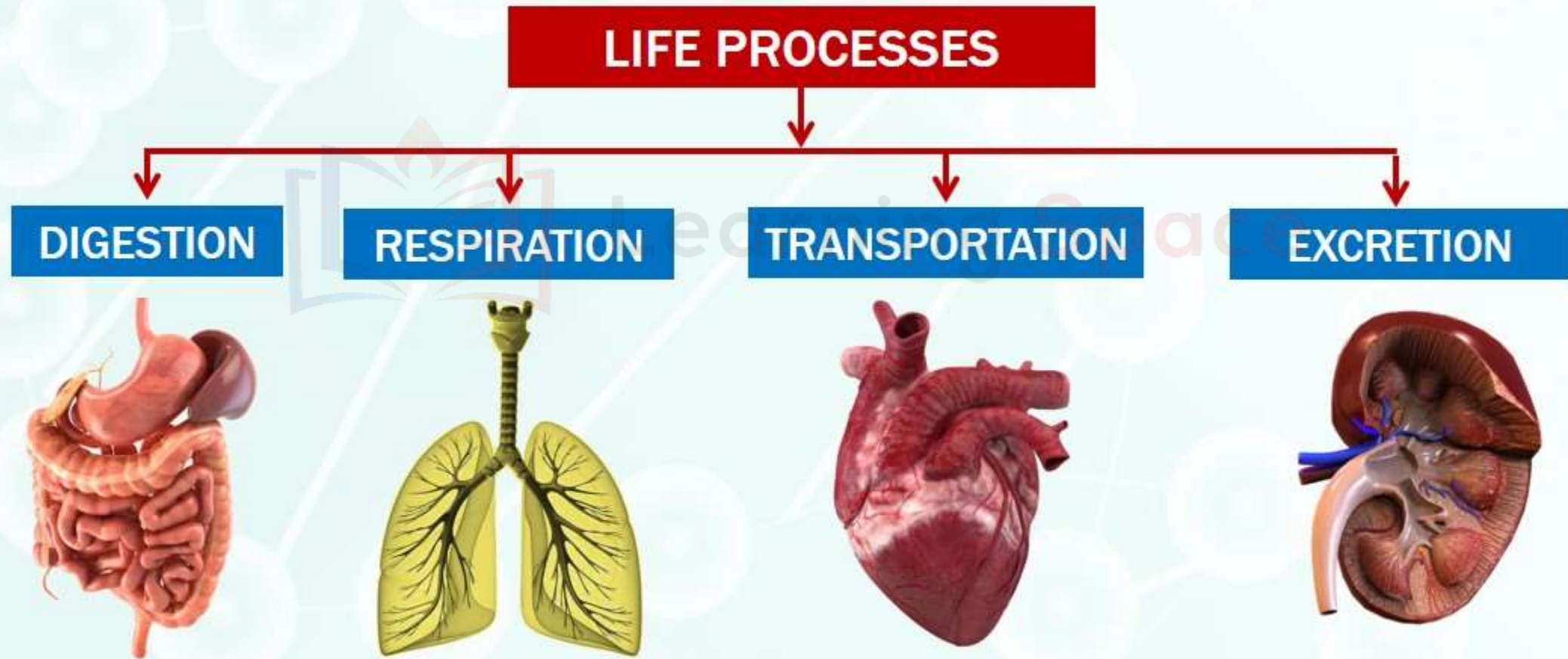
PYRUVATE



lactic acid

ATP





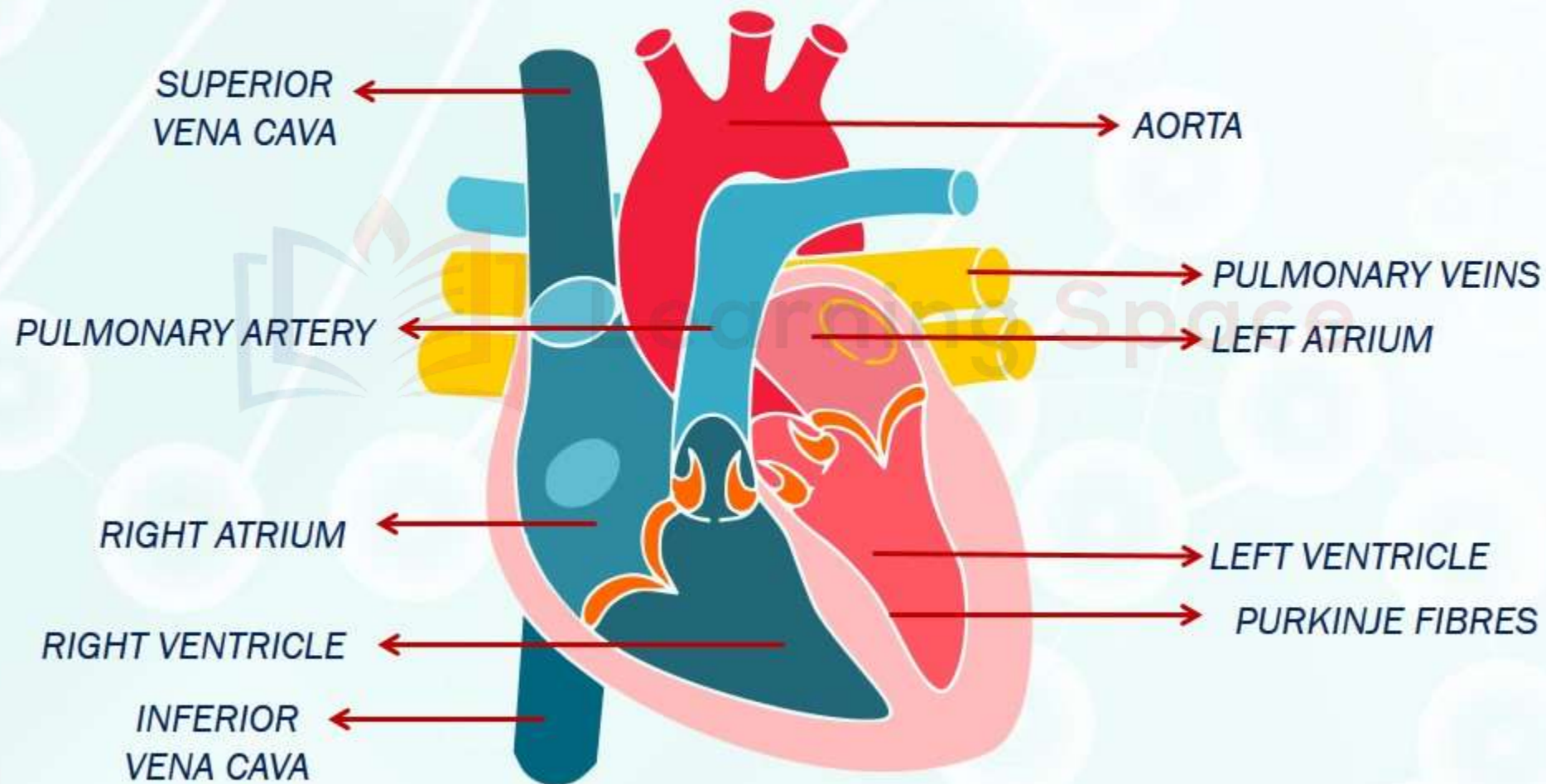
TRANSPORTATION



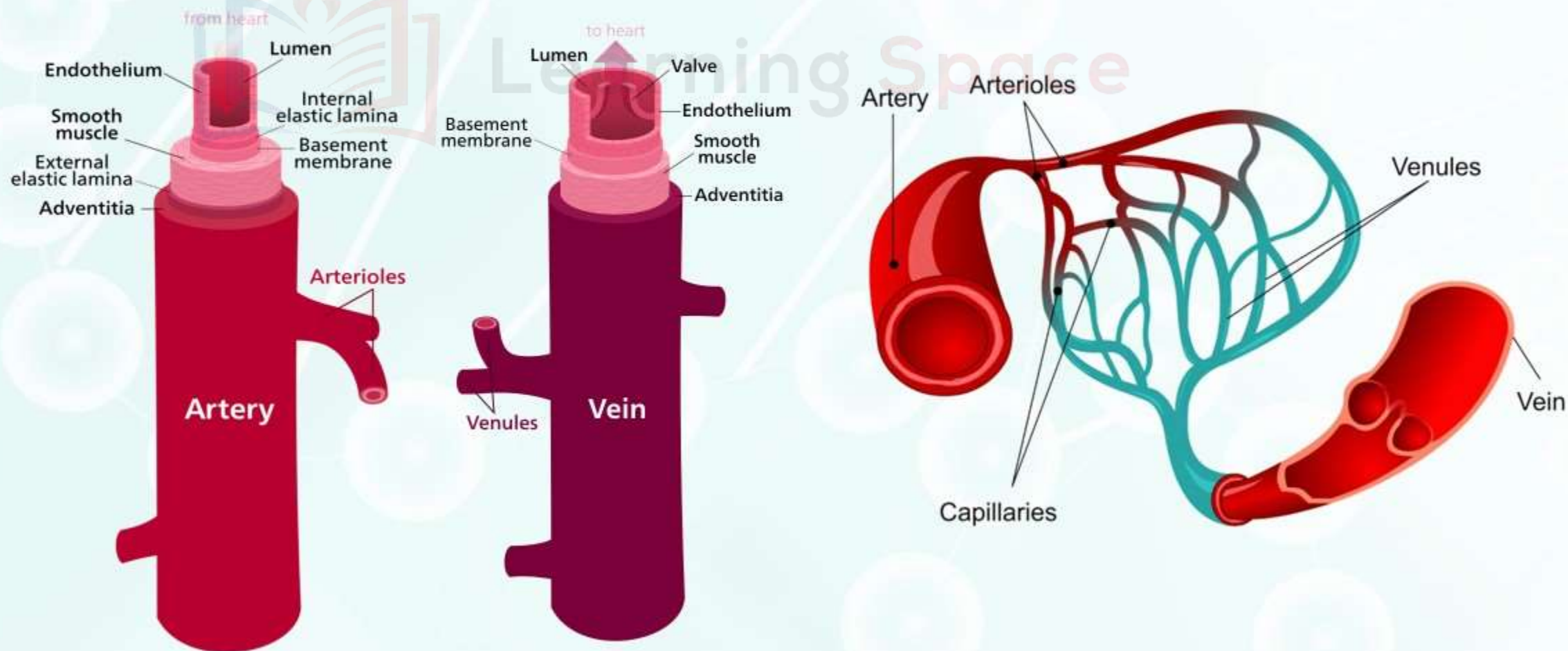
- Transportation system in humans consist of Heart and blood vessels.
- Blood flows through the blood vessels from the heart to all parts of the body.
- They transport digested food, oxygen to the cells and tissues of the body.
- CO_2 and cellular nitrogenous waste is also transported back to the heart and kidneys respectively.
- Together heart and the blood vessels are called the circulatory system.



THE CIRCULATORY SYSTEM IN HUMANS



THE BLOOD VESSELS – CARRIERS OF BLOOD



BLOOD VESSELS – DIFFERENCE BETWEEN THE TWO CLASSES

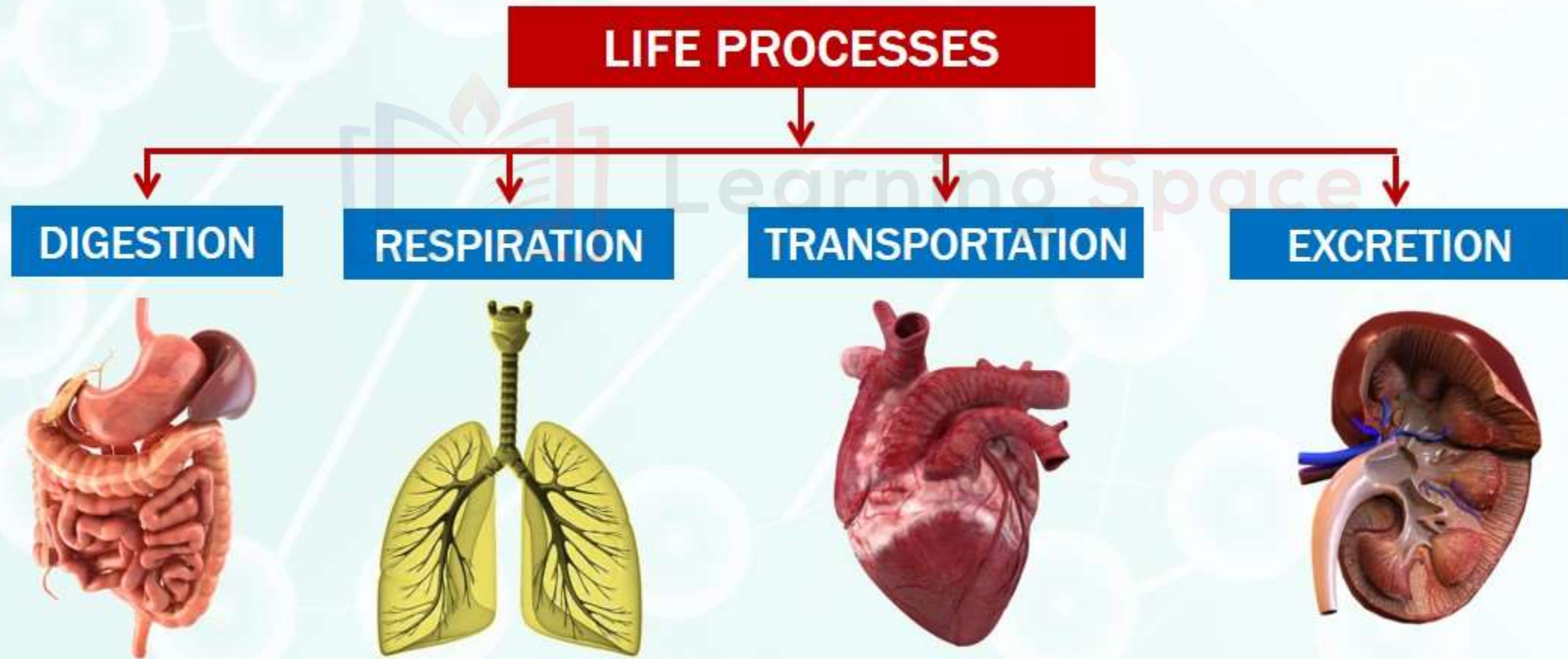
ARTERY

- Arteries are blood vessels that carry blood away from the heart.
- It has thick walls
- These are deep seated in the body
- No valves is present, the flow has pressure
- Carry oxygenated blood
- Exception – pulmonary artery

VEIN

- Veins are blood vessels that carry blood towards the heart.
- It has thin walls
- They are superficially located and can be seen through
- Valves are present because the blood may flow back
- Carry deoxygenated blood
- Exception – pulmonary vein





EXCRETION

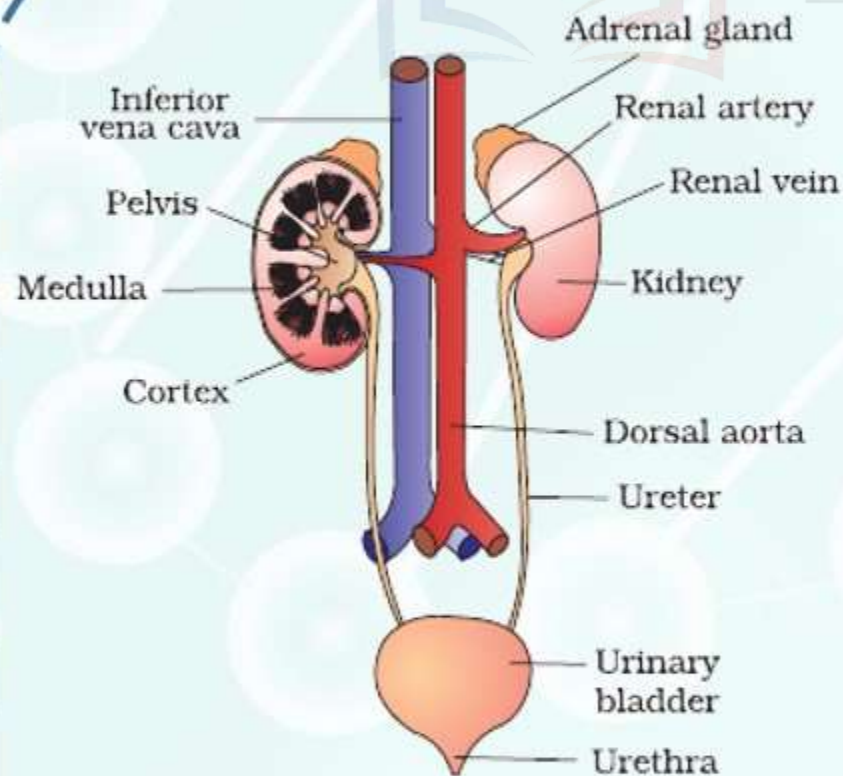


- The process of removal of harmful metabolic wastes from the body is called excretion.
- Our principal excretory organ is a pair of kidneys.
- The other organs include skin, liver and large intestine.
- In humans, the main excretory product is urea.
- The average human urine is 1 -2 litres ,that is churned out of almost 145 litres of blood cycling through the kidneys(GFR).

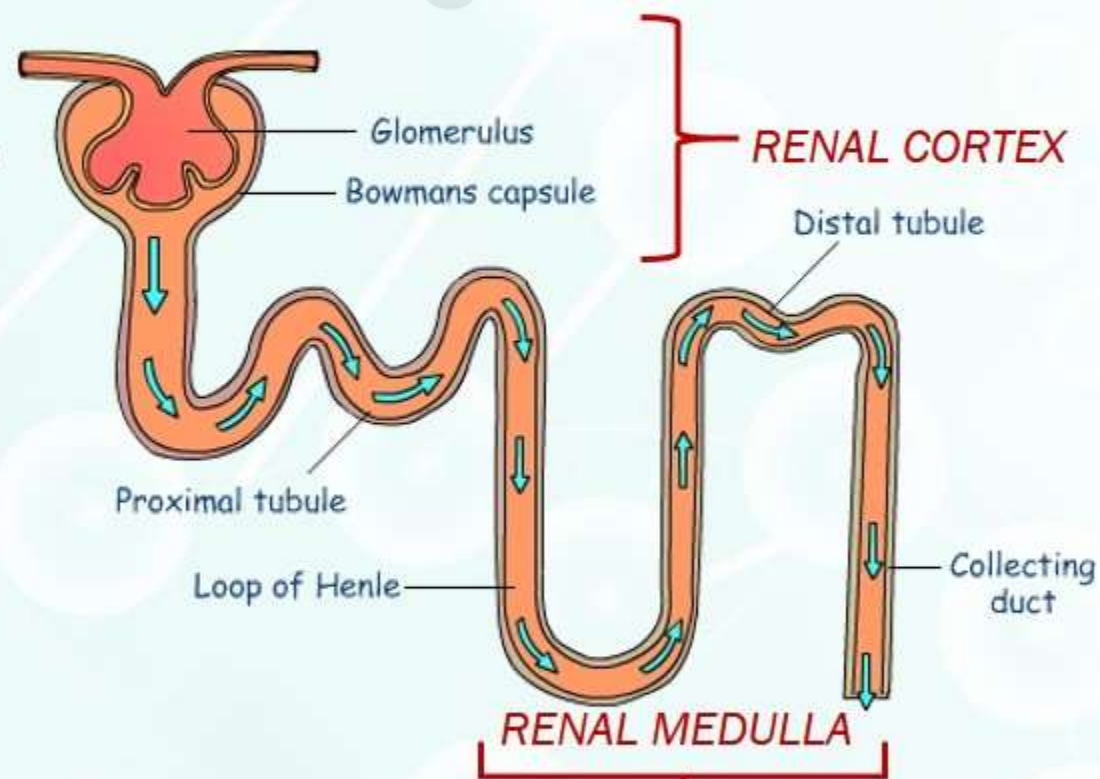
Reptiles and birds usually excrete uric acid in solid form



THE EXCRETORY ORGAN SYSTEM IN HUMANS



THE EXCRETORY SYSTEM - OVERVIEW



NEPHRON - THE FUNDAMENTAL UNIT



WE EXPRESS OUR
SINCERE THANKS
FOR VIEWING THIS VIDEO

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